

IBM Data Analysis Capstone Project

End-to-End Exploration from Data Collection to Interactive Dashboarding

Presented by

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Date

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OUTLINE

- I. Executive Summary
- II. Introduction
- III. Methodology
- IV. Results
 - Visualizations
 - Dashboards
- V. Discussion
 - Finding and Implications
- VI. Conclusion
- VII. Appendix

I. EXECUTIVE SUMMARY

Key Findings:

- **Current Technology Trends:** languages, databases, and platforms dominate current usage, with web frameworks being highly fragmented.
- **Future Technology Trends:** There is a clear shift in interest toward modern, high-growth languages and NoSQL/cloud-based databases and platforms.
- **Demographics:** The developer community is predominantly young, concentrated in a few major countries, with a high level of formal education.
- **Correlation Insights:** Compensation is strongly correlated with experience and specific skill sets, while job satisfaction is linked to remote work opportunities.
- Overall, the analysis highlights a fast-evolving tech landscape where continuous upskilling is essential.

II. INTRODUCTION

- **Purpose of the Report:**

This report presents an end-to-end data analysis project. The goal is to collect data from multiple sources (APIs, web scraping, and CSV files), clean and analyze it, and visualize key trends through charts and interactive dashboards in IBM Cognos.

- **Target Audience:**

This analysis is designed for tech recruiters, hiring managers, and aspiring developers who want to understand current and future trends in the technology market.

- **Value of the Analysis:**

The analysis provides valuable insights into in-demand skills, technology preferences, and demographics. It helps stakeholders make data-driven decisions about hiring, learning paths, and technology adoption.

III. Methodology

- **Data Sources:** CSV Files, APIs , Websites
- **Collection Methods :** Downloaded CSV file, Used API to get JSON data, Scraped websites with BeautifulSoup.
- **Data Wrangling Methods:**
 - Finding duplicates, visualizing them, and removing them.
 - Finding missing values, counting them, imputing them.
 - Normlizing data and visualizing the effect.
- **Data Analysis Methods:**
 - Exploratory Data Analysis:** Examined dataset structure, handled missing values, and performed summary statistics.
 - Distribution Analysis:** Analyzed data distribution using statistical summaries and visual methods.(data centered or skewed? spread wide or narrow?)
 - Outlier Detection:** Identified and removed outliers using statistical techniques.
 - **Correlation Analysis:** Explored relationships and correlations between key variables.
- **Data Visualization Method:**
 - Applied histograms, box plots, scatter plots, bubble plots, pie charts, stacked charts, line charts, and bar charts to represent insights.
- **Dashboarding:**
 - Create dashboards in IBM Cognos for Current Usage, Future Trends, and Demographics Bar chart, Column chart, Word cloud, Hierarchy bubble chart, Tree map chart, Pie chart, Map chart, Line chart, and Stacked bar chart.
- **Reporting:**
 - Present findings and recommendations in final report.

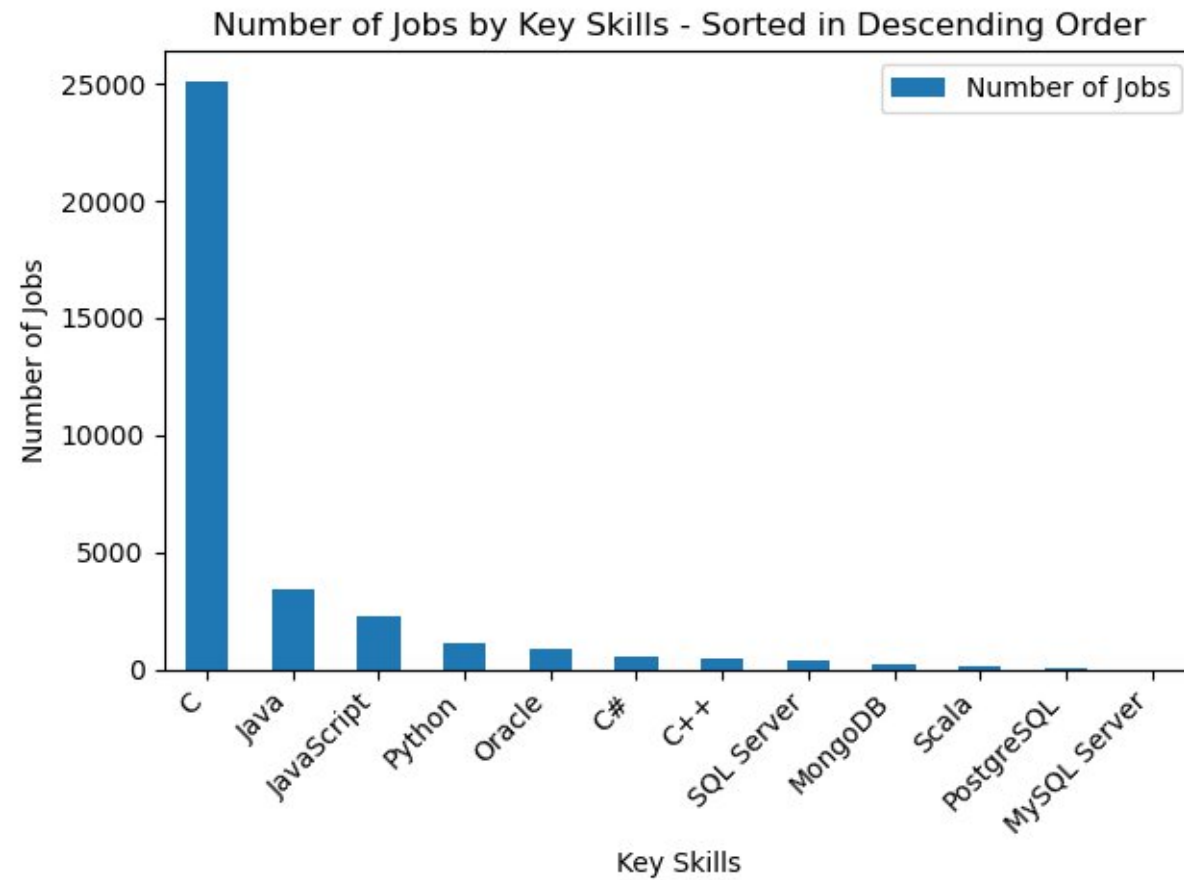
SECTION

IV. RESULTS

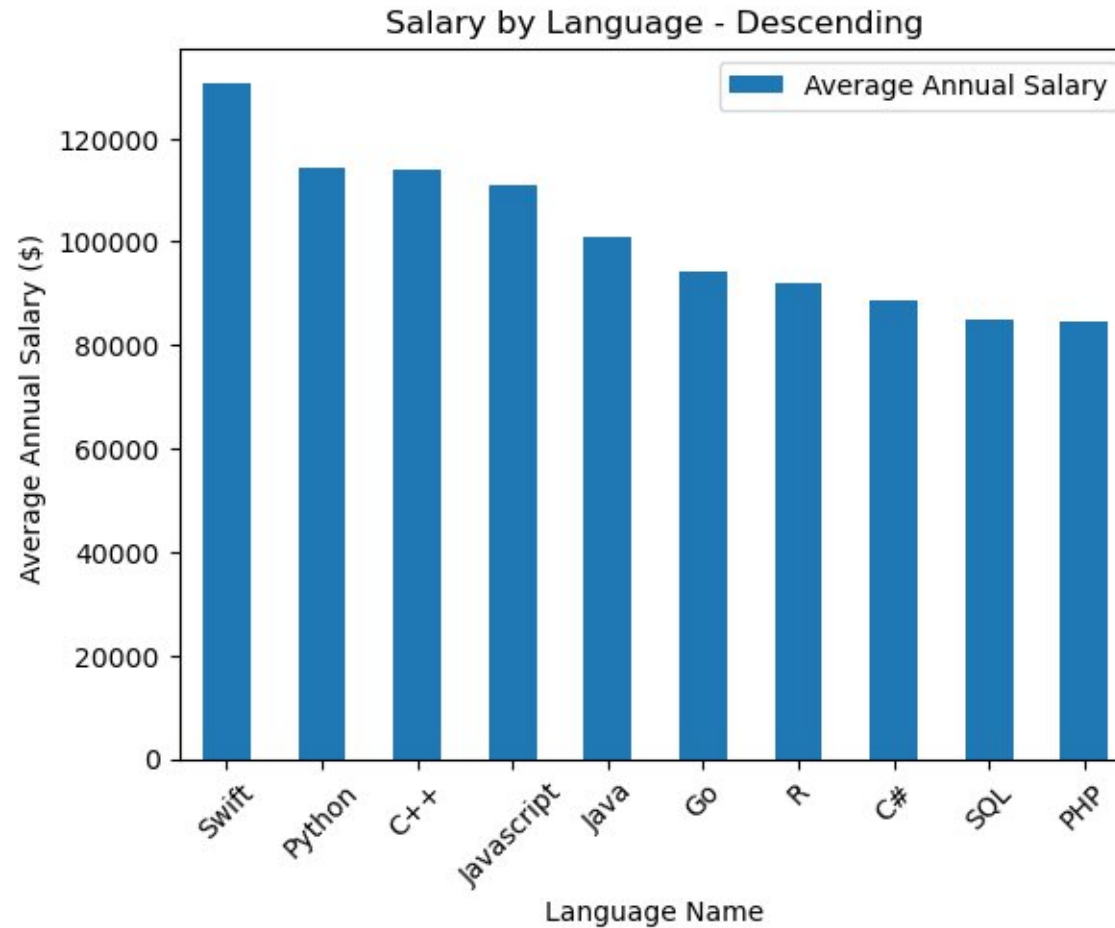
Presenting analysis from job postings, salary scraping, programming languages, and databases. Next slides.



A. JOB POSTINGS



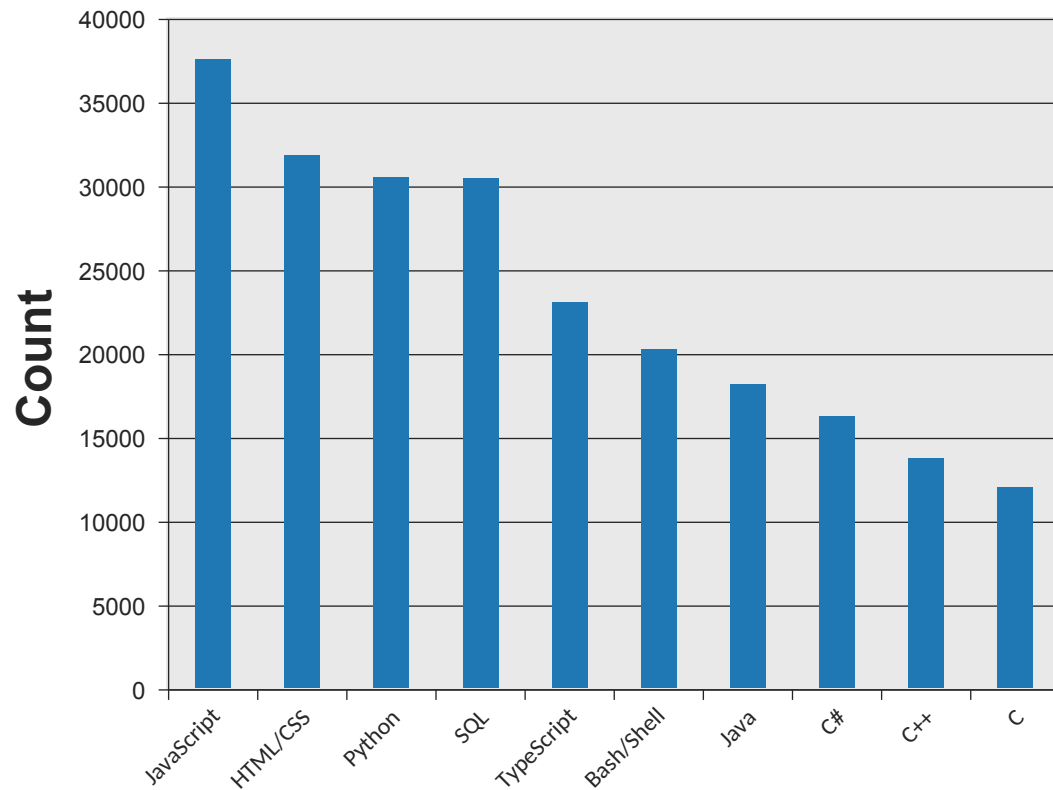
B. POPULAR LANGUAGES



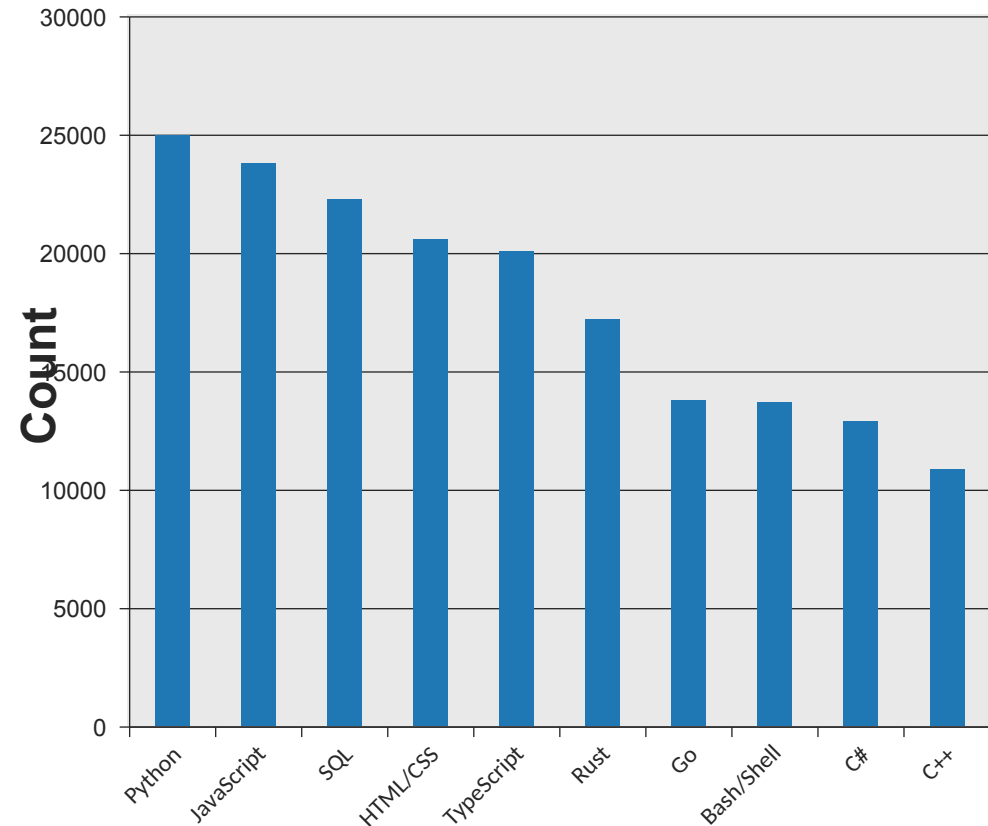
C. PROGRAMMING LANGUAGE TRENDS

Insight: JavaScript is the first language currently, Python leads the desire for next year . Interest in Rust & Go increased.

Top 10 Languages for this year



Top 10 Languages for next year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Key Findings

- JavaScript is #1 today (37.6k) but Python becomes #1 for next year (25k) - Python overtakes JS in desire.
- Web core remains strong: JavaScript, HTML/CSS, and TypeScript stay in Top 5 for both years.
- TypeScript steady growth (23k -> 20k) shows shift to typed JavaScript in large projects.
- Major Shift: Rust (#6 next year) and Go (#7 next year) enter Top 10, replacing Java and C. Rust +73% interest.
- Python + SQL dominance signals Data/AI boom - both stay in Top 3 for next year.
- Java drops out of Top 10 next year (18.2k -> out) - decline in enterprise preference among respondents.

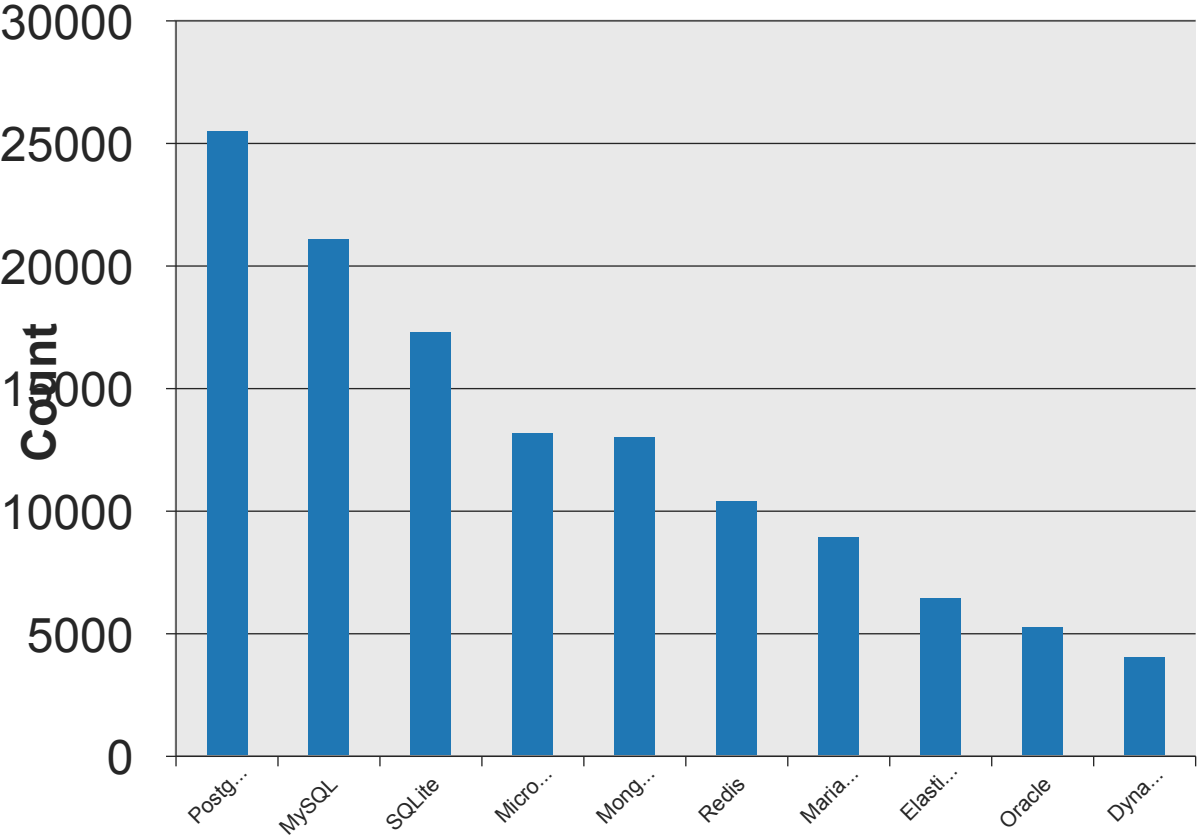
Implications & Recommendations

- For Job Seekers / Learners: Learn Python + SQL + JavaScript/TypeScript as the baseline stack. Add Rust or Go for high-value future roles in systems & cloud.
- For Companies / Hiring: High demand shift to Python (AI/Data) and Rust/Go (performance, cloud-native). Plan to upskill Java teams to these languages.
- For Education / Curriculum: Focus training on Python for AI, and TypeScript instead of plain JavaScript. Introduce Rust for system programming.
- For Business Strategy: Web technologies (JS/TS) are not going away, but investment in Python ecosystems (data, ML, automation) will give highest ROI next year.
- For Project Planning: If starting a new project next year, prioritize Python backend + TypeScript frontend, consider Go for microservices, Rust for performance-critical parts.

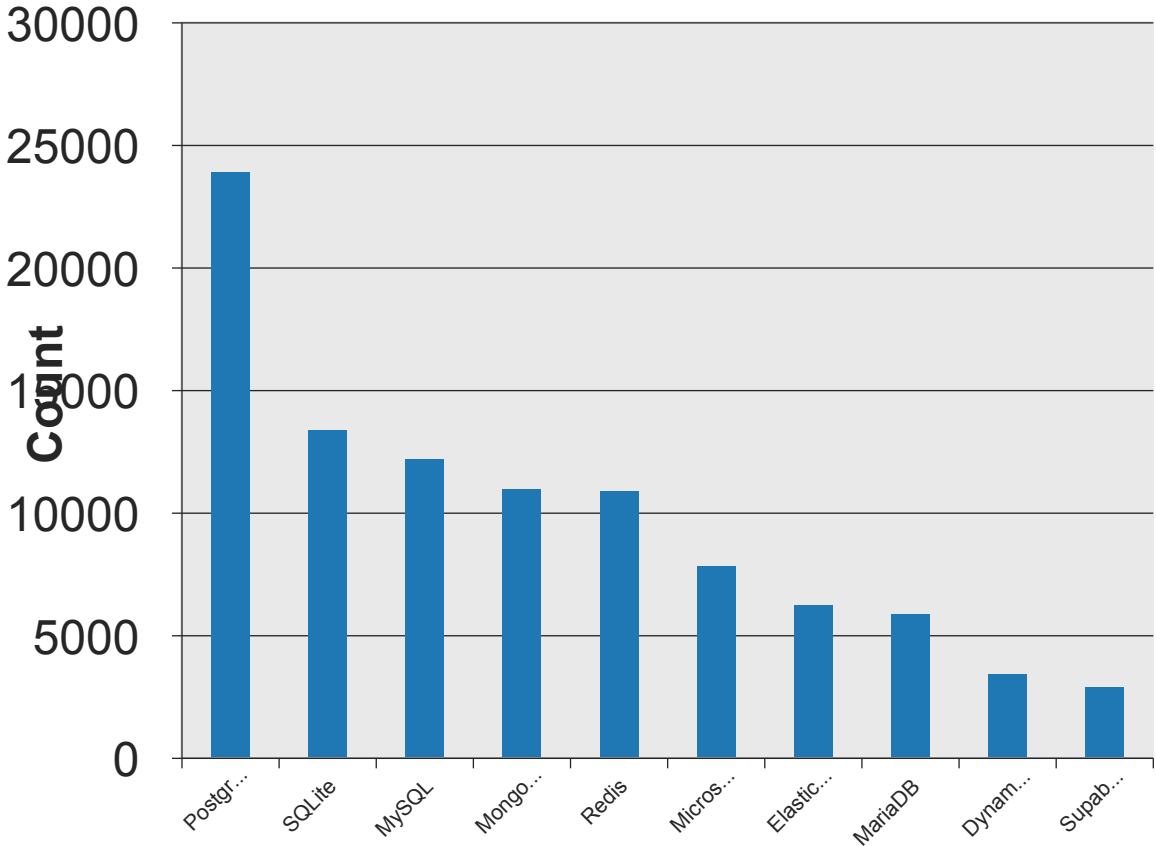
D. DATABASE TRENDS

Insight: PostgreSQL is the 1st in current and future demand, SQLite & MongoDB/Redis are growing, Oracle dropped out, Supabase became from theTop 10 next year.

Top 10 databases for the current year



Top 10 databases for the next year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Key Findings

1. PostgreSQL is clear leader: #1 both years (25.5k now, 23.9k next year) - 2x more than #2.
2. MySQL drops: From 2nd place to 3rd place in next year..
3. SQLite rises: From #3 to #2 for next year (13.4k) - light, embedded DB trend for mobile/edge.
4. NoSQL stays strong: MongoDB and Redis stay in Top 5 both years - demand for flexible & caching DBs.
5. Microsoft SQL Server declines: #4 to #6 (13.2k -> 7.8k) - shift away from enterprise proprietary.
6. New trend: Oracle drops out of Top 10 next year, Supabase (open-source Firebase) enters at #10.

Implications & Recommendations

- For Learners: Learn PostgreSQL as primary skill. Add SQLite for mobile/apps, and MongoDB/Redis for modern stacks.
- For Companies: Standardize on PostgreSQL - it dominates current and future. Reduce Oracle / MS SQL licensing costs.
- For Developers: NoSQL knowledge (MongoDB, Redis) is now mandatory alongside SQL - hybrid database skills needed.
- For Future Projects: Consider Supabase for rapid prototyping - it is entering Top 10, open-source alternative to Firebase.
- For Career: PostgreSQL + MongoDB + Redis combo is most future-proof. Oracle and MS SQL demand is declining.



DASHBOARD

IBM COGNOS ANALYTICS • 3 TABS

TAB 1

Current Technology Usage

TAB 2

Future Technology Trends

TAB 3

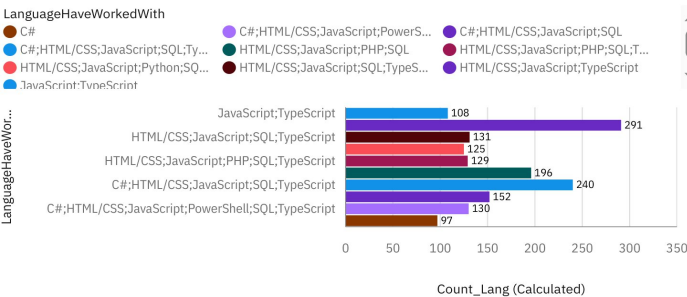
Demographics

DASHBOARD TAB 1: Current Technology Usage

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Current_Technology_Usage

Top 10 Languages WorkedWith

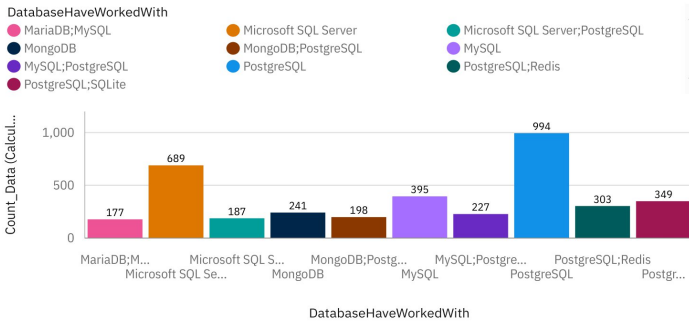


Top 10 Platforms WorkedWith

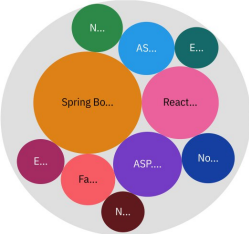
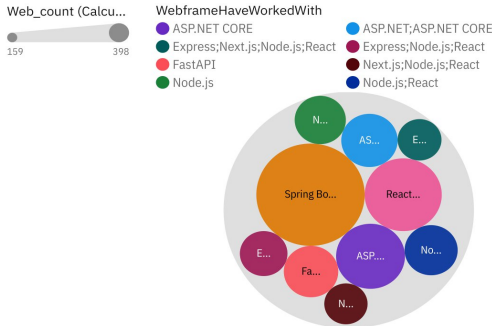


Current Technology Usage

Top 10 Databases WorkedWith



Top 10 Webframes WorkedWith



DASHBOARD INSIGHT

This Cognos dashboard shows CURRENT technology stacks actually used by developers across 4 areas: Languages, Databases, Platforms, and Web Frameworks. Data is based on Count of users who have worked with each technology.

SUMMARY OF KEY INSIGHTS

1. Languages: HTML/CSS;JS;TS combo #1 (291). Full-stack JS dominates current work.
2. Databases: PostgreSQL #1 (994) - double MySQL combo (689). PostgreSQL is the clear winner.
3. Platforms: AWS (2,391) is massive - 10x bigger than next platform. Azure #2.
4. WebFrames: Spring Boot is largest bubble, then React, ASP.NET CORE - Enterprise Java + React dominates.
5. Overall: Stack = TypeScript/JavaScript + PostgreSQL + AWS + Spring Boot/React.

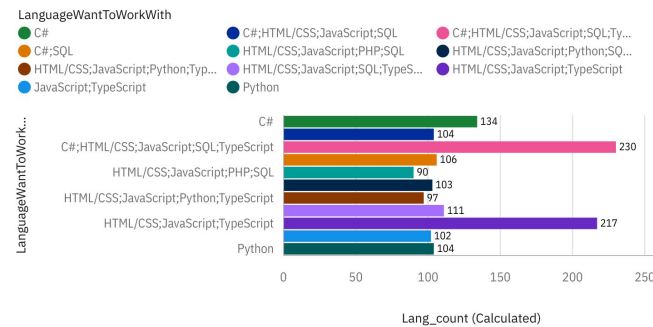
DASHBOARD TAB 2: Future Technology Trends

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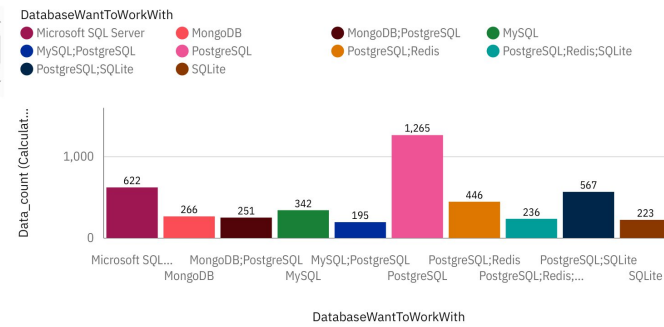
Future Technology Trend.

Future_Technology_Trend

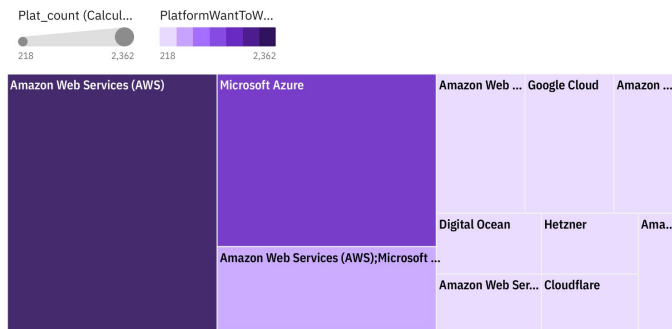
Top 10 Languages Want To Work With



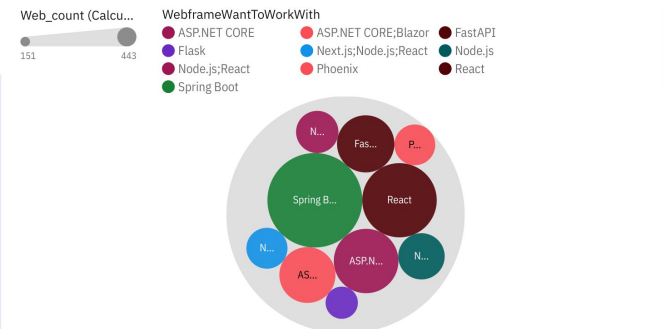
Top 10 Data Want to Work With



Top 10 Platforms Want to Work With



Top 10 Webframes Want to Work With



DASHBOARD INSIGHT

This dashboard shows FUTURE technology desires - what developers want to work with next year. Covers same 4 areas: Languages, Databases, Platforms, WebFrames, based on future interest counts.

SUMMARIZ OF KEY INSIGHTS

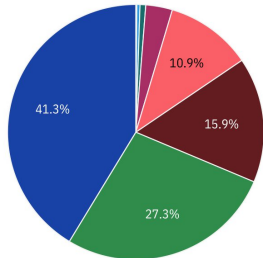
1. Languages: C#;HTML/CSS/JS/SQL/TS #1 (230), HTML/CSS/JS/TS #2 (217) - TypeScript full-stack most desired.
2. Databases: PostgreSQL #1 (1,265) huge lead over MS SQL (622). PostgreSQL + Redis combo rising (446).
3. Platforms: AWS #1 (2,362) still leads future, Microsoft Azure #2, AWS+Google Cloud combo growing.
4. Webframes: Spring Boot biggest bubble (future #1), then React, ASP.NET CORE - Spring Boot + React stays top.
5. Overall: Future = TypeScript-focused full-stack + PostgreSQL + AWS + Spring Boot/React.

DASHBOARD TAB 3: Demographics

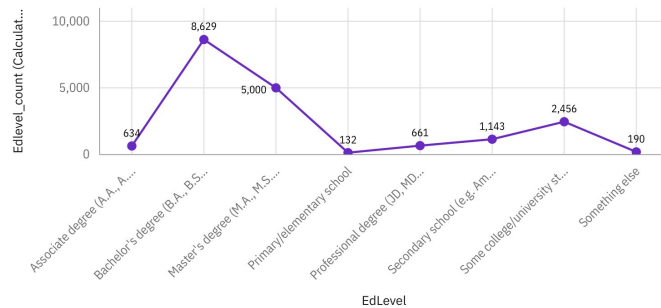
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Demographics

Age
● Prefer not to say
● 45-54 years old
● 65 years or older
● Under 18 years old
● 18-24 years old
● 35-44 years old
● 55-64 years old
● 25-34 years old

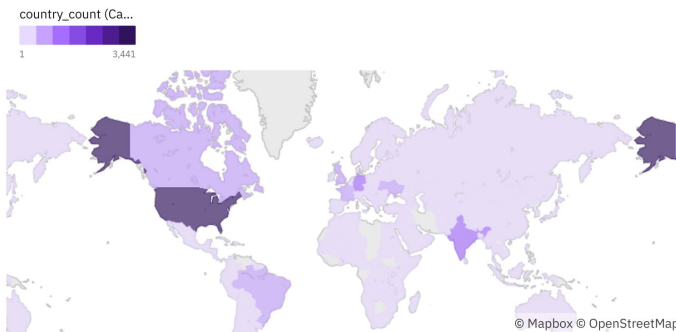


Edlevel_count by EdLevel

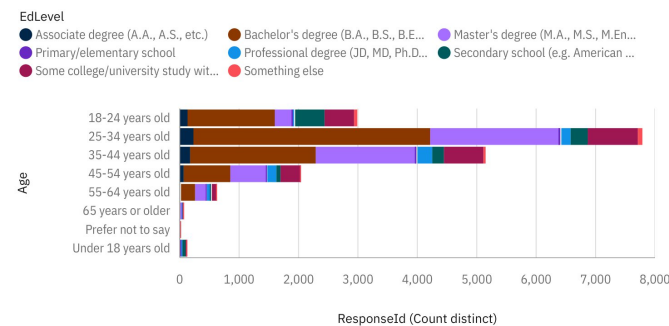


Demographics

country_count by Country



ResponseId by Age colored by EdLevel



DASHBOARD INSIGHT

This dashboard shows WHO the respondents are: Age distribution, Country distribution, Education Level, and Age colored by Education. Based on Stack Overflow survey demographic data.

SUMMARIZY OF KEY INSIGHTS

1. Age: 25-34 years old is largest group (41.3%), then 35-44 (27.3%) - developers are mostly young professionals.
2. Country: US dominates (3,441) - darkest purple on map - followed by India. Survey is US-centric.
3. Education: Bachelor's degree highest (8,629), Master's #2 (5,000) - majority have university degrees.
4. Age vs EdLevel: 25-34 age group holds most Bachelor's (brown) + Master's (light purple) degrees.
5. Overall: Audience = Young (25-34), US-based, Bachelor-educated developers.

V. DISCUSSION

1. Current vs Future Stack

- Current = JavaScript/TypeScript, PostgreSQL (994), AWS (2,391), Spring Boot/React.
- Future = Same leaders stay: TypeScript full-stack (230), PostgreSQL (1,265), AWS (2,362), Spring Boot still #1.
- Key shift: From Microsoft SQL (689) to PostgreSQL + Redis (446) - move to open-source + caching.
- Language trend: HTML/CSS/JS/PHP drops (129) → TypeScript-focused stacks rise.
- Implication: Stack is stable - learn PostgreSQL, TypeScript, AWS, React/Spring.

2. Who is Using These Technologies?

- Age: 25-34 is core audience (41.3%) + 35-44 (27.3%) = 68% are 25-44 young professionals.
- Education: 8,629 Bachelor's + 5,000 Master's = Highly educated workforce.
- Country: US (3,441) dominates - data is US-centric, but India rising in future trends.
- Age x Education: 25-34 holds most Bachelor's & Master's degrees - entry to mid-level devs.
- Implication: Tools like AWS, PostgreSQL, TypeScript target educated young pros - need easy learning curve.

3. Overall Strategic Insights

- Open-source wins: PostgreSQL beats Microsoft SQL & Oracle in current AND future - companies saving license costs.
- Cloud consolidation: AWS is 2,391 vs Azure - AWS is 10x leader now and future (2,362) - multi-cloud with AWS core.
- Full-stack TypeScript: Future desire moves to C#;HTML/CSS/JS/SQL/TS (230) - TypeScript is the bridge.
- Java + React remains enterprise standard: Spring Boot biggest in both Current and Future webframes.
- Recommendation: For career - Master PostgreSQL + TypeScript + AWS + React/Spring Boot. This stack serves the main audience (25-44, Bachelor's, US) and is future-proof.

VI. OVERALL FINDINGS & IMPLICATIONS

Findings

Finding 1 - PostgreSQL and AWS Dominate Current and Future: PostgreSQL is #1 Current (994) and #1 Future (1,265), AWS is 2,391 Current and 2,362 Future. Both are 2x-10x bigger than competitors. Microsoft SQL Server drops from 689 to 622, Oracle drops out of Top 10.

Finding 2 - TypeScript Full-Stack and Spring Boot + React is Standard: Current top is HTML/CSS/JS/SQL/TypeScript (291), Future top is C#,HTML/CSS/JS/SQL/TypeScript (230) and HTML/CSS/JS/TypeScript (217). Spring Boot is biggest webframe Current (398) and Future (443), React is second. Full-stack TypeScript is now required.

Finding 3 - Audience is Young, Educated, US-Centric: 41.3% are 25-34 years old + 27.3% 35-44 = 68% are young professionals. Education: 8,629 Bachelor's degree, 5,000 Master's. Country: US 3,441 dominates, with India and UK next. This explains preference for modern open-source tools like PostgreSQL + Redis (446 future) and Supabase entering Top 10.

Implications

Implication 1 - For Developers and Job Seekers: Learn PostgreSQL + TypeScript + AWS + Spring Boot/React. This is the most future-proof stack for the main audience (25-34, Bachelor's). Avoid investing heavily in Oracle or pure PHP stacks (drops from 129 to 90). Add Redis and Supabase for modern projects.

Implication 2 - For Companies and Startups: Cut costs by migrating from proprietary databases (Oracle, Microsoft SQL Server) to open-source PostgreSQL and Supabase. Standardize cloud on AWS as primary with Azure secondary. Use TypeScript to reduce bugs and Spring Boot + React for enterprise-grade applications that are #1 now and in future.

Implication 3 - For Industry and Education: The industry is shifting to open-source, cloud-native, and TypeScript-based development. Educational programs should focus on PostgreSQL, AWS Cloud, TypeScript full-stack, and using Cognos dashboards for data analysis. Companies targeting young, educated US developers must provide easy-to-learn, cloud-based tools.

VII. CONCLUSION

- Successfully executed a full data analysis lifecycle, from collecting data via APIs and web scraping to data wrangling, exploratory analysis, and visualization.
- Identified dominant technologies in current use and emerging trends showing high interest in modern open-source and cloud technologies.
- Analyzed demographic distribution and its correlation with education, age, and compensation.
- Built interactive dashboards in IBM Cognos using multiple chart types — Bar, Column, Word Cloud, Bubble, Tree Map, Pie, Map, Line, and Stacked Bar — to communicate insights effectively.
- The project demonstrates practical skills in data cleaning (handling missing values, duplicates, normalization) and storytelling with data.

Next Steps: Expand the dataset, automate dashboard updates, and include deeper predictive analysis for future skill demand.

VIII. APPENDIX

Relevant additional charts or tables created during analysis:

- Dashboard Links:

- Current Technology Usage Dashboard
- Future Technology Trends Dashboard
- Demographics Dashboard

Reference: IBM Cognos Analytics

- Tools & Libraries Used:

- Python: Pandas, Matplotlib, Requests, BeautifulSoup
- IBM Cognos Analytics, GitHub